

FINA2204 Tutorial 2: Futures markets

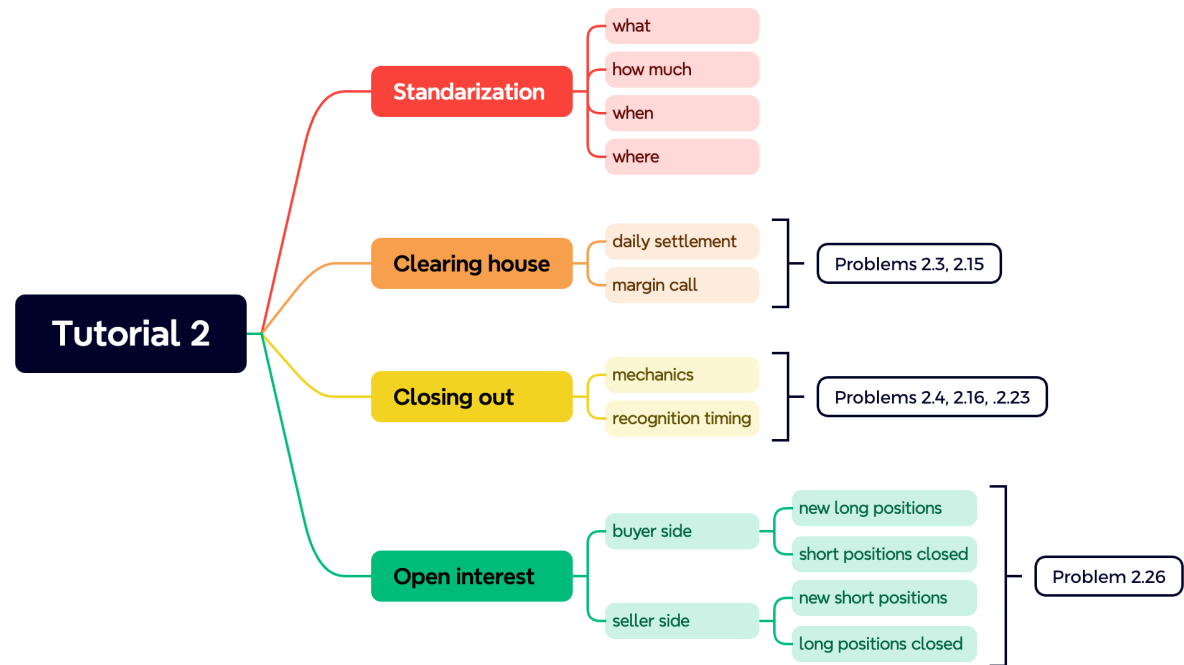
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Agenda

- Problem 2.3
- Problem 2.4
- Problem 2.15
- Problem 2.16
- Problem 2.23
- Problem 2.26



Presented with xmind

Recap: futures contract

What is a futures contract?

Key idea

A **futures contract** is a standardised, exchange-traded agreement to buy or sell a specified asset, in a specified quantity and quality, at a price agreed today, for delivery (or cash settlement) on a specified future date.

- Key features: (a) **Standardization**: contract terms are fixed, making contracts of the same type interchangeable; (b) **Clearing house**: it manages margin and daily settlement.

Recap: futures contract

Standardization – CME groups

- The world's largest and most diverse derivative marketplace.
- Offers standardized contracts across major asset classes.
- It specifies **what** is traded, **how much** is traded, and **when** and **where** delivery takes place.
- Example: WTI crude oil futures



CRUDE OIL FUTURES - CONTRACT SPECS

Source: <https://www.cmegroup.com/markets/energy/crude-oil/light-sweet-crude.contractSpecs.html>

CONTRACT UNIT	1,000 barrels
PRICE QUOTATION	U.S. dollars and cents per barrel
TRADING HOURS	CME Globex: Sunday - Friday 5:00 p.m. - 4:00 p.m. CT with a 60-minute break each day beginning at 4:00 p.m. CT TAS: Sunday - Friday 5:00 p.m. - 1:30 p.m. CT CME ClearPort: Sunday 5:00 p.m. - Friday 4:00 p.m. CT with no reporting Monday - Thursday from 4:00 p.m. - 5:00 p.m. CT
MINIMUM PRICE FLUCTUATION	0.01 per barrel = \$10.00 TAS: Zero or +/- 10 ticks in the minimum tick increment of the outright
PRODUCT CODE	CME Globex: CL CME ClearPort: CL Clearing: CL TAS: CLT TAM: "CLS","CLC","CLL"
LISTED CONTRACTS 7/8/2026	Monthly contracts listed for the current year and the next 10 calendar years and 2 additional contract months. List monthly contracts for a new calendar year and 2 additional contract months following the termination of trading in the December contract of the current year. FINA2204 - Tutorial 2

OVERVIEW

QUOTES

SETTLEMENTS

VOLUME & OI

SPECS

MARGINS

CALENDAR

FUTURES

OPTIONS -

VENDOR CODES	Quote Vendor Symbols Listing ↗
DELIVERY PROCEDURE	Delivery shall be made free-on-board ("F.O.B.") at any pipeline or storage facility in Cushing, Oklahoma with pipeline access to Enterprise, Cushing storage, Enbridge, Cushing storage or Plains, Cushing storage. Delivery shall be made in accordance with all applicable Federal executive orders and all applicable Federal, State and local laws and regulations. At buyer's option, delivery shall be made by any of the following methods: (1) by interfacility transfer ("pumpover") into a designated pipeline or storage facility with access to seller's incoming pipeline or storage facility; (2) by in-line (or in-system) transfer, or book-out of title to the buyer; or (3) if the seller agrees to such transfer and if the facility used by the seller allows for such transfer, without physical movement of product, by in-tank transfer of title to the buyer.
DELIVERY PERIOD	(A) Delivery shall take place no earlier than the first calendar day of the delivery month and no later than the last calendar day of the delivery month. (B) It is the short's obligation to ensure that its crude oil receipts, including each specific foreign crude oil stream, if applicable, are available to begin flowing ratably in Cushing, Oklahoma by the first day of the delivery month, in accord with generally accepted pipeline scheduling practices. (C) Transfer of title-The seller shall give the buyer pipeline ticket, any other quantitative certificates and all appropriate documents upon receipt of payment. The seller shall provide preliminary confirmation of title transfer at the time of delivery by telex or other appropriate form of documentation.
GRADE AND QUALITY	Please see rulebook chapter 200 FINA2204 - Tutorial 2 5

Non-standardized
assets destroy
liquidity!

200101.A. Domestic Crudes

1. Deliverable Crude Streams

Blends of West Texas Intermediate ("WTI") type light sweet crude streams are only deliverable if such blends constitute a pipeline's designated "common stream" shipment which meets the grade and quality specifications for domestic crude. Enterprise Products Partners L.P. (including any successor in such capacity, "Enterprise") and Enbridge Pipeline (Ozark) LLC's (including any successor in such capacity, "Enbridge") and Plains Marketing, L.P.'s (a subsidiary of Plains including any successor in such capacity, "Plains") Common Domestic Sweet ("DSW") Streams that meet quality specifications in Sections 101.A.2.- 12. of this rule are deliverable as Domestic Crude.

2. Sulfur: 0.42% or less by weight as determined by ASTM Standard D-4294, or its latest revision;

3. Gravity: Not less than 37 degrees American Petroleum Institute ("API"), nor more than 42 degrees API as determined by ASTM Standard D-287, or its latest revision;

4. Viscosity: Maximum 60 Saybolt Universal Seconds at 100 degrees Fahrenheit as measured by ASTM Standard D-445 and as calculated for Saybolt Seconds by ASTM Standard D-2161;

5. Reid vapor pressure equivalent: Less than 9.5 pounds per square inch at 100 degrees Fahrenheit, as determined by ASTM Standard D-6377, or its latest revision;

6. Basic Sediment, water and other impurities: Less than 1% as determined by ASTM D-96-88 or D-4007, or their latest revisions;

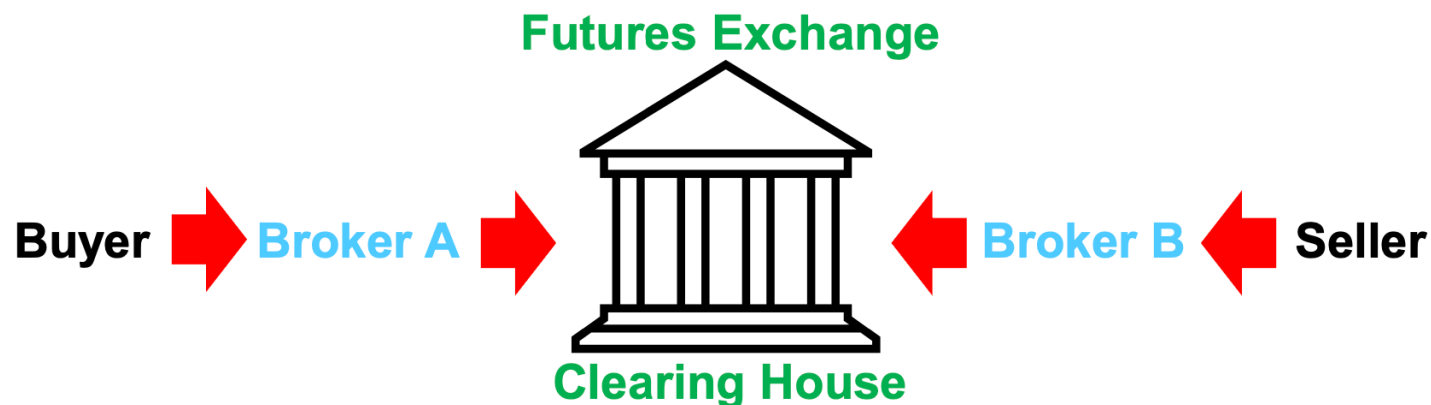
7. Pour Point: Not to exceed 50 degrees Fahrenheit as determined by ASTM Standard D-97;

Source: NYMEX Rulebook,
<https://www.cmegroup.com/rulebook/NYMEX/2/200.pdf>

Recap: futures contract

Clearing house

- Stands between the buyer and seller in every futures trade.
- Becomes the buyer to every seller and the seller to every buyer.
- It eliminates counterparty credit risk!



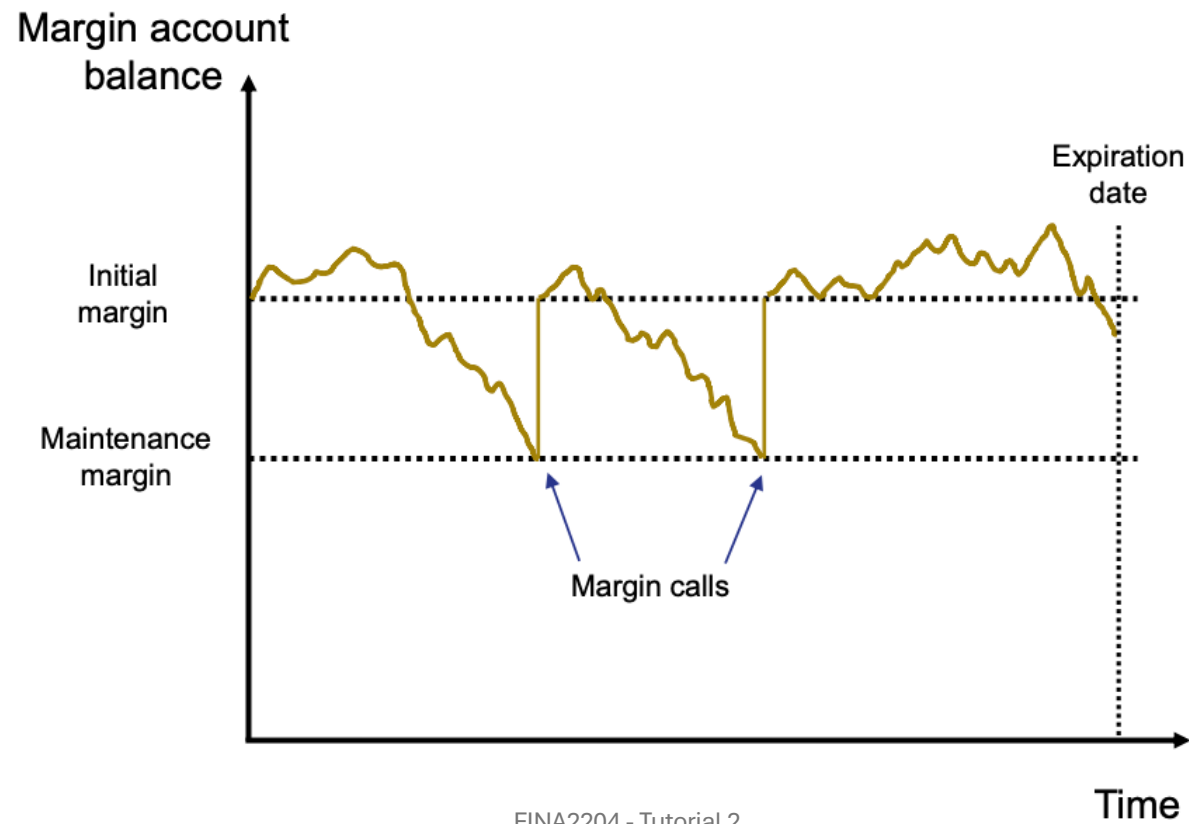
Recap: futures contract

- To manage counterparty exposure, the clearing house requires a **margin** account as collateral to absorb potential defaults.
- Initial margin: deposit required to open the position.
- Maintenance margin (~75%) is the minimum allowed balance. If the balance falls below it, a **margin call** restores it to the initial margin.
- Fail to pay: the broker closes you out.
- Balance above initial margin can be **withdrawn**.

Recap: futures contract

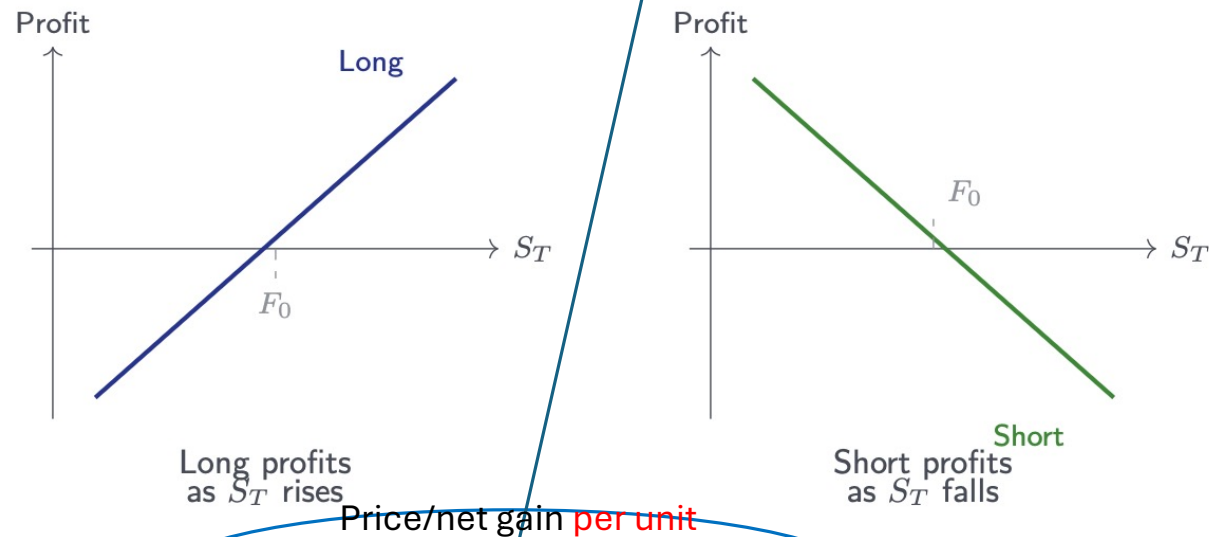
- **Daily settlement:** gains and losses are settled based on the new futures price each day.
- Economically, the contract is rewritten each day to reset trader debt to zero.
- Together, daily settlement keeps the debt tiny, and the margin account ensures there is always enough cash to pay it.

Recap: futures contract



Notional value at entry=future price per unit * contract size * number of contracts
Total gain/loss=net gain per unit * contract size * number of contracts

Recap: futures contract



- Profit to the long = $S_T - F_0$; to the short = $F_0 - S_T$. They sum to zero: futures are a **zero-sum** transfer.
- The lines are **straight** and **unbounded** both ways, the defining shape of a *linear* (delta-one) payoff. Options, from Week 6, bend these lines.

Recap: futures contract

SETUP

- Contract size (suppose): 100 ounces per contract
- Number of contracts: 2
- Notional value at entry is: $1,250 \times 100 \times 2 = \$250,000$
- **Leverage = $250,000 / 12,000 = 20.83$!**

Table 2.1 Operation of margin account for a long position in two gold futures contracts. The initial margin is \$6,000 per contract, or \$12,000 in total; the maintenance margin is \$4,500 per contract, or \$9,000 in total. The contract is entered into on Day 1 at \$1,250 and closed out on Day 16 at \$1,226.90

	Day	Trade price (\$)	Settlement price (\$)	Daily gain (\$)	Cumulative gain (\$)	Margin account balance (\$)	Margin call (\$)
(1,241-1,250)*100*2	1	1,250.00				12,000	
→0.72% (1,241/1,250)	1		1,241.00	-1,800	-1,800	10,200	
fall in the gold price	2		1,238.30	-540	-2,340	9,660	
produces a 15%	3		1,244.60	1,260	-1,080	10,920	
(1,800/12,000) loss on	4		1,241.30	-660	-1,740	10,260	
the initial margin!	5		1,240.10	-240	-1,980	10,020	
	6		1,236.20	-780	-2,760	9,240	<\$9,000; Margin call!
	7		1,229.90	-1,260	-4,020	7,980	4,020
	8		1,230.80	180	-3,840	12,180	
	9		1,225.40	-1,080	-4,920	11,100	
	10		1,228.10	540	-4,380	11,640	
	11		1,211.00	-3,420	-7,800	8,220	3,780
	12		1,211.00	0	-7,800	12,000	
	13		1,214.30	660	-7,140	12,660	
	14		1,216.10	360	-6,780	13,020	
	15		1,223.00	1,380	-5,400	14,400	
	16	1,226.90		780	-4,620	15,180	

Source: textbook, p.48

Problem 2.3

- *Suppose that you enter into a short futures contract to sell July silver for \$27.20 per ounce. The size of the contract is 5,000 ounces. The initial margin is \$4,000, and the maintenance margin is \$3,000.*
- *What change in the futures price will lead to a margin call?*
- *What happens if you do not meet the margin call?*

Answer 2.3

Setup

- Initial margin: \$4,000
- Maintenance margin: \$3,000
- Number of contracts: 1 short future
- Contract size: 5,000 ounces
- Trade price (F_0): \$27.20
- Settle price (S_T): ?

Answer 2.3

- There will be a margin call when \$1,000 has been lost from the margin account.
- This will occur when the price of silver increases by $1000/5000 = \$0.20$.
- The price of silver must therefore rise to \$27.40 per ounce for there to be a margin call.
- If the margin call is not met, your broker closes your position.

Answer 2.3

$$\text{Payoff to the short per unit} = F_0 - S_T$$

$$\text{Total Profit} = (F_0 - S_T) \times \text{Contract Size} \times \text{Number of Contracts}$$

- Let $S_T = X$,
- $\text{Payoff to the short per unit} = \$27.20 - X$
- $\text{Total Profit} = (\$27.20 - X) \times 5,000 \times 1 = \$3,000 - \$4,000 = -\$1,000$
- $27.20 - X = \frac{-1,000}{5,000} = -\0.2
- $X = \$27.20 + \$0.2 = \$27.40$

Problem 2.15

- *At the end of one day a clearinghouse member is long 100 contracts and the settlement price is \$50,000 per contract. The original margin is \$2,000 per contract.*
- *On the following day the member becomes responsible for clearing an additional 20 long contracts, entered into at a price of \$51,000 per contract. The settlement price at the end of this day is \$50,200.*
- *How much does the member have to add to its margin account with the exchange clearinghouse?*

Answer 2.15

- The clearing house member is required to provide $20 \times \$2,000 = \$40,000$ as initial margin for the new contracts.
- There is a profit of $(50,200 - 50,000) \times 100 = \$20,000$ on the existing contracts.
- There is also a profit of $(50,200 - 51,000) \times 20 = -\$16,000$ on the new contracts.
- The member must therefore add $\$40,000 - (\$20,000 - \$16,000) = \$36,000$ to the margin account.

Recap: closing out

- A trader can either leave the initial position open and take delivery of the underlying asset, or 'close out' the position by entering an offsetting trade.
- Most positions (>95%) are offset before maturity, as participants prefer to close out rather than take delivery.
- For some contracts (e.g., stock index futures), delivery is not permitted; open positions are settled in cash at the spot price on the expiration date.

Q: Why do most participants close out before maturity?

Recap: closing out

- **Traders seek price exposure rather than physical delivery!**
 1. Speculators and arbitrageurs seek profits from price movements, rather than owning the physical asset.
 2. Hedgers usually continue using their regular supply chains.
 - Futures delivery imposes strict requirements concerning location, grade, timing, storage, and transportation
 - The physical transaction and the futures transaction are therefore handled separately.

Recap: closing out

Derivation: closing position

Close a futures position by taking an equal and opposite position in the same contract. No delivery is required.

- Original long: buy at F_1 , and then sell at F_2 .
- View at maturity T :

$$\text{Long payoff} = S_T - F_1$$

$$\text{Offsetting short payoff} = F_2 - S_T$$

$$\text{Net gain for the original long} = (\cancel{S_T} - F_1) + (F_2 - \cancel{S_T}) = F_2 - F_1$$

- For an original short, the net gain is $-(F_2 - F_1)$.

Problem 2.16

- *On 1 July 2013, a Japanese company enters into a forward contract to buy \$1 million with yen on 1 January 2014. On 1 September 2013, it enters into a forward contract to sell \$1 million on 1 January 2014.*
- *Describe the profit or loss the company will make in dollars as a function of the forward exchange rates on 1 July 2013 and 1 September 2013.*

Note: F_1, F_2 : Forward exchange rates on 1 July 2013 and 1 September 2013; S_T : Spot rate on 1 January 2014; All exchange rates are measured as yen per dollar.

Answer 2.16

$$\text{Payoff to the long} = S_T - F_0$$

- The payoff from the first contract is $(S - F_1)$ million yen.

$$\text{Payoff to the short} = -(S_T - F_0)$$

- the payoff from the second contract is $-(S_T - F_2)$ million yen.
- The total payoff is therefore $(S_T - F_1) - (S_T - F_2) = (F_2 - F_1)$ million yen.
- By entering the second contract, the company effectively locked in their profit or loss on September 1st, completely immunizing themselves from whatever the spot rate does in January.

Recap: closing out

Recognition timing

- While the economic gain ($F_2 - F_1$) is standard, the accounting and tax timing depends entirely on **the purpose of the trade**.
 1. Hedging: tax rules generally allow you to delay recognizing the futures gains or losses until the underlying hedged transaction is completed, matching them up in the same financial year.
 2. Speculation (mark-to-market basis): you must recognize gains and losses in the exact year they occur, even if you haven't closed the position yet.

Worked example: hedging vs speculating

In May a firm buys one October greasy wool futures at \$9.50/kg (2,500 kg). The price of the futures contract is at \$9.56 at 30 June, and the position is closed in October at \$9.58. Financial year ends 30 June.

Solution · recognition timing

As a hedge (matched to the October wool purchase):

- FY (to 30 June): **\$0** recognised, the gain is deferred to the hedged transaction.
- Following FY: $(9.58 - 9.50) \times 2,500 = \textbf{\$200}$ recognised in full.

As speculation (marked to market each year):

- FY (to 30 June): $(9.56 - 9.50) \times 2,500 = \textbf{\$150}$.
- Following FY: $(9.58 - 9.56) \times 2,500 = \textbf{\$50}$.

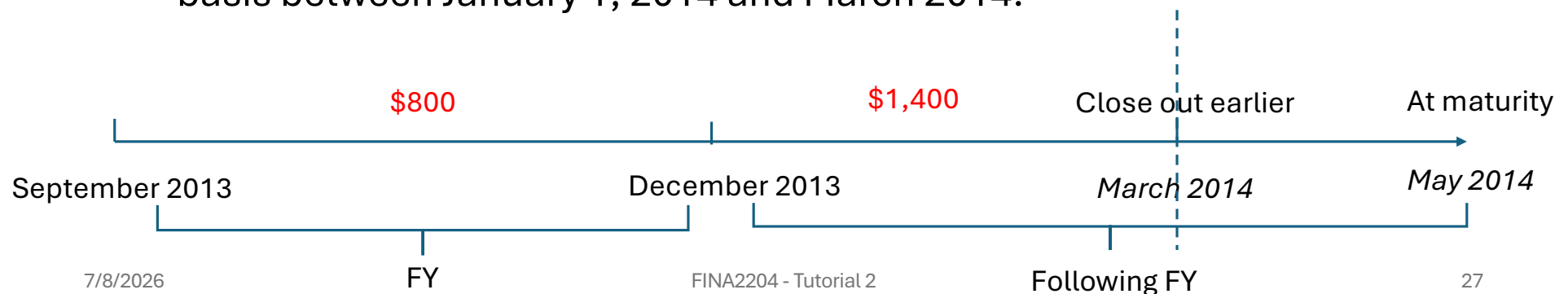
Total gain is \$200 either way; only the *timing* differs.

Problem 2.4

- *Suppose that in September 2013 a company takes a long position in a contract on May 2014 crude oil futures. It closes out its position in March 2014. The futures price (per barrel) is \$88.30 when it enters into the contract, \$90.50 when it closes out the position, and \$89.10 at the end of December 2013. One contract is for the delivery of 1,000 barrels.*
- *What is the company's profit?*
- *When is it realised? How is it taxed if it is (a) a hedger, and (b) a speculator? Assume that the company has a 31 December year end.*

Answer 2.4

- The net gain is $(S - F_1) - (S - F_2) = (F_2 - F_1) = \$90.50 - \$88.30 = \2.20 (*per barrel*)
- The total profit is $\$2.20 \times 1,000 = \$2,200$
 - Of this, $(\$89.10 - \$88.30) \times 1,000 = \$800$ is realized on a day-by-day basis between September 2013 and December 2013.
 - A further $(\$90.50 - \$89.10) \times 1,000 = \$1,400$ is realized on a day-by-day basis between January 1, 2014 and March 2014.



Answer 2.4

- A hedger would be taxed on the whole profit of \$2,200 in 2014.
- A speculator would be taxed on \$800 in 2013 and \$1,400 in 2014.

Problem 2.23

- *Suppose that on October 24, 2013, a company sells one April 2014 live-cattle futures contract. It closes out its position on January 21, 2014.*
- *The futures price (per pound) is 91.20 cents when it enters into the contract, 88.30 cents when it closes out the position and 88.80 cents at the end of December 2013. One contract is for the delivery of 40,000 pounds of cattle.*
- *What is the profit? How is it taxed if the company is (a) a hedger and (b) a speculator? Assume that the company has a December 31 year end.*

Answer 2.23

- The payoff is $-(S - F_1) + (S - F_2) = (F_1 - F_2) = \$0.9120 - \$0.8830 = \0.029 (*per pound*)
- The total profit is $\$0.029 \times 40,000 = \1160
- Of this, $(\$0.9120 - \$0.8880) \times 40,000 = \960 is realized on a day-by-day basis between October 2013 and December 2013.
- A further $(\$0.8880 - \$0.8830) \times 40,000 = \200 is realized on a day-by-day basis between January 1, 2014 and April 2014.

Answer 2.23

- A hedger would be taxed on the whole profit of \$1,160 in 2014.
- A speculator would be taxed on \$960 in 2013 and \$200 in 2014.

Recap: open interest

Open interest vs. volume

- Open interest is the number of outstanding contracts, equal to total long positions or total short positions.
- Volume is the number of contracts traded during a day and may exceed open interest if contracts are traded repeatedly.
- Every transaction increases volume by 1.
- Open interest rises only when both sides open new positions and falls when both sides close existing positions.

Recap: open interest

- Every trade has one buyer and one seller, and each side can either (i) open a new position or (ii) close an existing one. This creates four possible scenarios:
 - a. Both open new positions: a new long and a new short are created.
 - b. Buyer opens, seller closes: one long holder replaces another.
 - c. Buyer closes, seller opens: one short holder replaces another.
 - d. Both close: One long and one short disappear.

Recap: open interest

Let:

- *buyer side*

1. $NL = \text{new long positions} = a + b$

2. $CS = \text{closed short positions} = c + d$

- *Seller side*

1. $NS = \text{new short positions} = a + c$

2. $CL = \text{closed long positions} = b + d$

Recap: open interest

- Trade volume

$$V = NL + CS = NS + CL = a + b + c + d$$

- Change in open interest (OI)

- Total long position side

$$\Delta OI = NL - CL = (a + \cancel{b}) - (\cancel{b} + d) = a - d$$

Or



- Total short position side

$$\Delta OI = NS - CS = (\cancel{a} + \cancel{c}) - (\cancel{c} + d) = a - d$$

Problem 2.26

- *Explain what is meant by open interest.*
- *Why does the open interest usually decline during the month preceding the delivery month?*
- *On a particular day, there were 2,000 trades in a particular futures contract. This means that there were 2,000 buyers (going long) and 2,000 sellers (going short).*
- *Of the 2,000 buyers, 1,400 were closing out positions and 600 were entering into new positions. Of the 2,000 sellers, 1,200 were closing out positions and 800 were entering into new positions. What is the impact of the day's trading on open interest?*

Answer 2.26

- Open interest is the number of contracts outstanding.
- Many traders close out their positions just before the delivery month is reached. This is why the open interest declines during the month preceding the delivery month.
- The open interest **went down by 600**. We can see this in two ways. First, 1,400 shorts closed out and there were 800 new shorts. Second, 1,200 longs closed out and there were 600 new longs.

Answer 2.26

- *buyer side*

1. $NL = \text{new long positions} = 600$

2. $CS = \text{short positions closed} = 1,200$

$$\Delta OI = NL - CS = 600 - 1,200 = -600$$

- *Seller side*

1. $NS = \text{new short positions} = 800$

2. $CL = \text{long positions closed} = 1,400$

$$\Delta OI = NS - CL = 800 - 1,400 = -600$$